

6.4.1 The nuclear atom

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) alpha-particle scattering experiment; evidence of a small charged nucleus	
(b) simple nuclear model of the atom; protons, neutrons and electrons	
(c) relative sizes of atom and nucleus	M0.4, M1.4
(d) proton number; nucleon number; isotopes; notation ${}^A_Z\text{X}$ for the representation of nuclei	
(e) strong nuclear force; short-range nature of the force; attractive to about 3 fm and repulsive below about 0.5 fm	1 fm = 10^{-15} m
(f) radius of nuclei; $R = r_0 A^{1/3}$ where r_0 is a constant and A is the nucleon number	
(g) mean densities of atoms and nuclei.	